

Homework 6

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This study mainly investigates how the average number of responses to a newspaper advertisement varies by the day of the week it's published, the section in which it's published or a combination of these two. To explore whether these two factors result in significant differences in the number of responses, as well as to determine if there is a potential interaction between these two factors, this report conducts a two-way ANOVA. The primary research questions to address in this report are whether the average number of responses differs by day of the week, by newspaper section, and whether the effect of one factor depends on the level of the other.

Before conducting the ANOVA analysis, we tested two basic assumptions to ensure the validity of the model. First, we tested for normality. Since each cell contains only four observations, normality must be evaluated at the cell level rather than based on the overall sample size. We verified that all skewness ratios were less than 2 in magnitude (with a maximum of 0.64), indicating that the normality assumption is satisfied. Next, we tested for homogeneity of variances by using Levene's test. The p-value obtained was 0.975, which is well above the 0.05 significance level. Therefore, we fail to reject the null hypothesis, indicating that the assumption of equal variances is met. Given that both assumptions have been met and the sample sizes are balanced, we can proceed directly with the ANOVA analysis using the existing dataset.

To visualize the relationship between the average number of responses and different workdays and sections, we created the following graph using R:

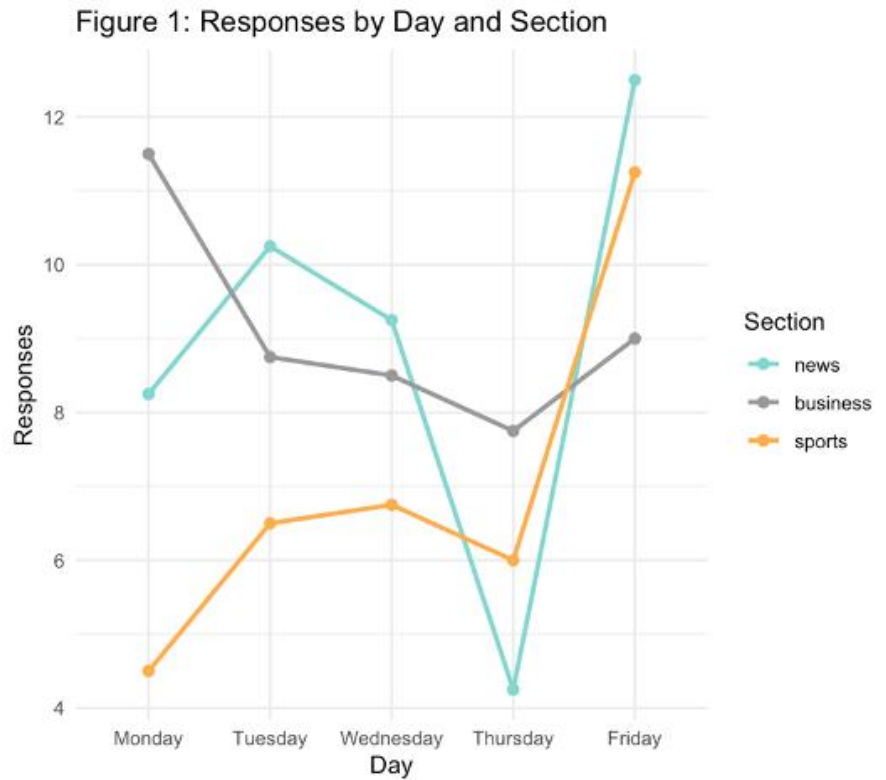


Figure 1 shows that responses in the news section peaked on Friday but dropped sharply on Thursday. In contrast, the Business section performed best on Monday, followed by a general downward trend. The Sports section consistently received fewer responses than the other two sections from Monday through Thursday, but its response rate improved significantly by Friday. The frequent intersections of the lines in the graph suggest strong interactions between the variables. This indicates that the effect of day on responses depends on the section in which the advertisement is placed. For example, the lowest response occurs on Thursday for the news section but on Monday for the sports section. In addition, the Business section reaches its highest average response on Monday, whereas both the News and Sports sections peak on Friday. These results suggest that advertisers cannot only rely on a single day or section. Instead, they must consider both factors comprehensively to maximize consumer engagement.

A two-way ANOVA was conducted to examine the effects of day of the week and newspaper section on the number of responses, as well as their interaction. The results revealed a statistically significant interaction between day and section since the p-value is less than 0.001. This indicates that the effect of day on the number of responses depends on the section in which the advertisement is placed. The partial eta squared for the interaction was 0.63, indicating that after controlling for the main effects of day and section, approximately 63% of the variance in responses can be explained by the interaction between day and section. This represents a large effect size according to Cohen's guidelines since the value is greater than 0.14.

When controlling for day of the week, the effect of the section on responses varies across days. On Monday, the business section received significantly more responses than both the news and sports sections, while the news section also received significantly more responses than the sports section. This indicates that the business section performed best on Monday, followed by news, with sports performing the worst. On both Tuesday and Wednesday, the news section received significantly more responses than the sports section, while no significant difference was observed between the other section pairs. On Thursday, the business section received significantly more responses than the news section, while no significant differences were observed between the other section pairs. On Friday, the news section received significantly more responses than the business section, while no significant difference was observed between the other section pairs. The table below reports mean differences for statistically significant comparisons only. For instance, "Monday Business - News +3.25" refers to the fact that the average number of responses in the business section is 3.25 higher than in the news section on Monday.

Day	Comparison	Difference (Responses)
Monday	Business – News	+3.25
	News – Sports	+3.75
	Business – Sports	+7.00
Tuesday	News – Sports	+3.75
Wednesday	News – Sports	+2.50
Thursday	Business – News	+3.50
Friday	News – Business	+3.50

When controlling for sections, the effect of the day on responses varies across sections. Within the news section, Thursday had significantly lower responses compared to all other four days while Friday had significantly higher responses than Monday, Wednesday and Thursday but did not differ significantly from Tuesday. This indicates Thursday performed the worst within the news section while Friday tended to produce the relatively strong responses. Within the business section, responses on Monday were significantly higher than those on Wednesday and Thursday, while no other significant differences were observed between the remaining days. This suggests that Monday tends to have higher responses compared to some days, although it is not significantly different from all other days. Within the sports section, responses on Friday were significantly higher than those on all other days, while no significant differences were observed among the remaining days. This indicates that Friday performed best in the sports section. The table below reports mean differences for statistically significant comparisons only. For instance, “News Monday - Thursday +4.00” indicates that the average number of responses on Monday is 4.00 higher than on Thursday in the news section.

Section	Comparison	Difference (Responses)
News	Monday – Thursday	+4.00
	Tuesday – Thursday	+6.00
	Wednesday – Thursday	+5.00
	Friday – Monday	+4.25
	Friday – Wednesday	+3.25
	Friday – Thursday	+8.25
Business	Monday – Wednesday	+3.00
	Monday – Thursday	+3.75
Sports	Friday – Monday	+6.75
	Friday – Tuesday	+4.75
	Friday – Wednesday	+4.50
	Friday – Thursday	+5.25

In conclusion, the results of the two-way ANOVA demonstrate that both the day of the week and the newspaper section significantly influence the number of responses to advertisements. The significant interaction indicates that the effectiveness of advertisement placement depends on the combination of both day and section. The Business section tends to achieve higher responses on Monday, while the News and Sports sections tend to perform best on Friday. Additionally, the magnitude of differences between sections varies across days, and the pattern of responses across days differs by section. These findings suggest that advertisers should not rely on a single factor when making decisions. Instead, they should consider both the timing and placement of advertisements simultaneously in order to maximize audience engagement and improve the effectiveness of their advertisements.